

Telecom Users Analysis Report

Overview

The data based on the Telecom_users dataset mainly categorises into three types of information.

Customer Information which contains:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents

Service Information:

- Tenure
- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

Account Information:

- Contract
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges

Summaries

Summary of Non-Senior Subscribers



Figure 1: Total non-senior subscribers for both male and female

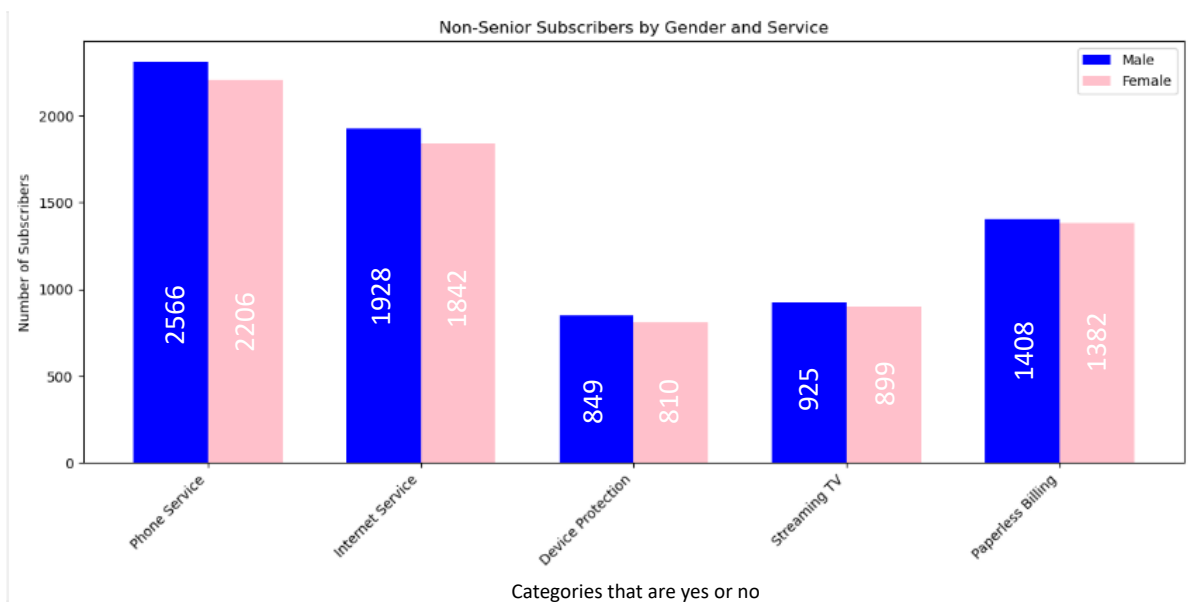


Figure 2: This table shows the non-senior subscribers based on gender and categories that are primarily answered yes or no.

The Table displays primarily services with the addition of paperless billing column. This is mainly because the input is either yes or no. Therefore, all subscribers number are those with input of yes.

From the table above we can see that the distribution of subscribers based on gender is very similar against the service categories. We can also observe that overall, there are generally more male subscribers than female subscribers for each service.

Summaries

Summary of Senior Subscribers

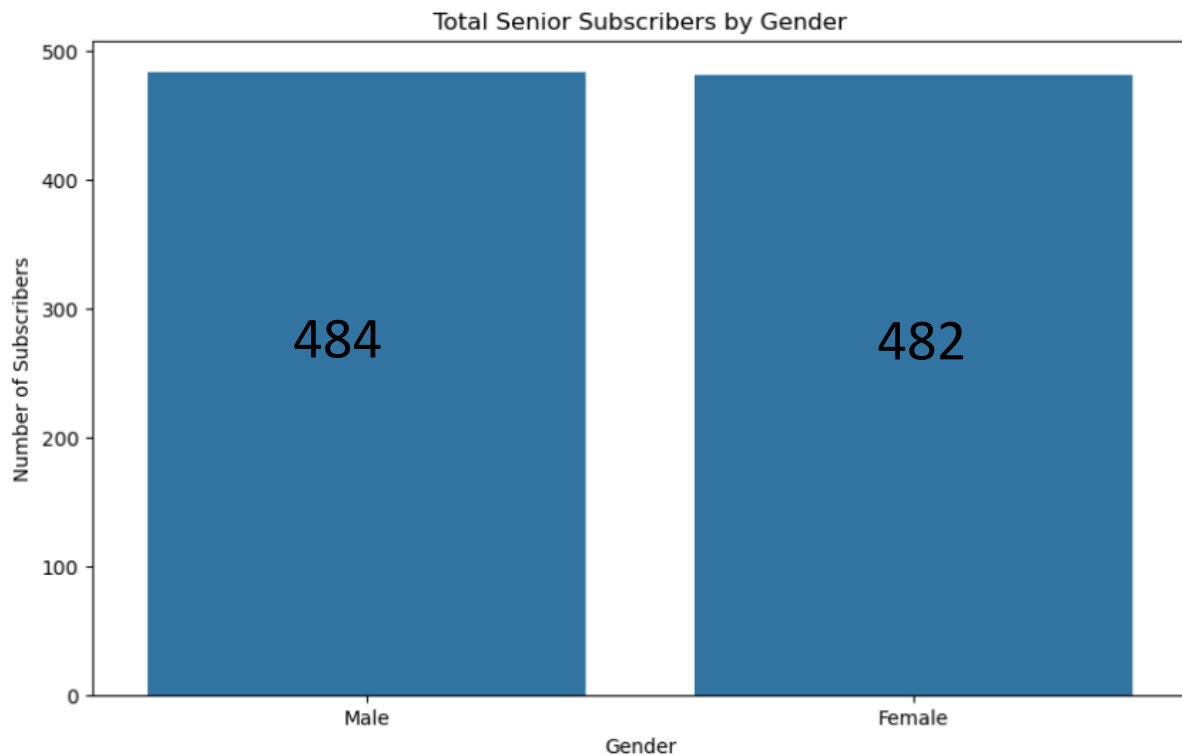


Figure 3: Total number of senior subscribers based on gender

Based on the numbers shown in the graph, we can see that there are significantly fewer senior subscribers compared to non-senior subscribers by more than 5 times

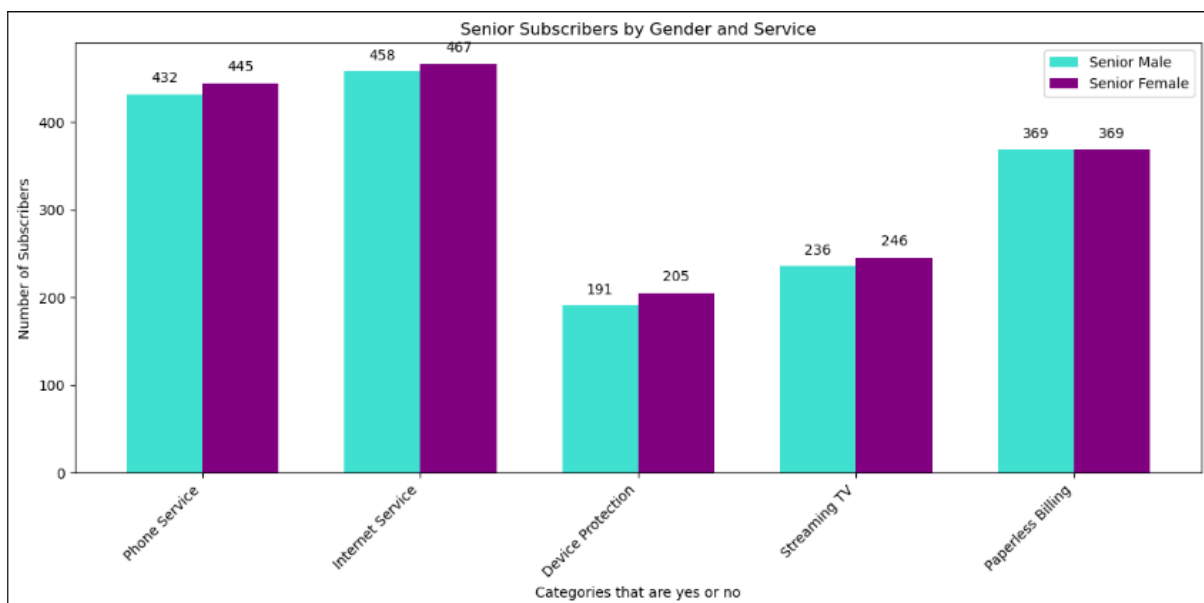


Figure 4: This table shows the senior subscribers based on gender and categories that are primarily answered yes or no.

Summaries

When looking at senior subscribers we can see different trends. The overall distribution of the services aside from internet services which is the highest in count is the most similar to non-senior subscribers.

On the other hand, for senior subscribers there is in general more female subscribers for each service which is the complete opposite for non-seniors where there is generally more male than female subscribers for each service.

From the two tables we can observe the preferences in services for different demography based on gender and those with and without senior citizenship

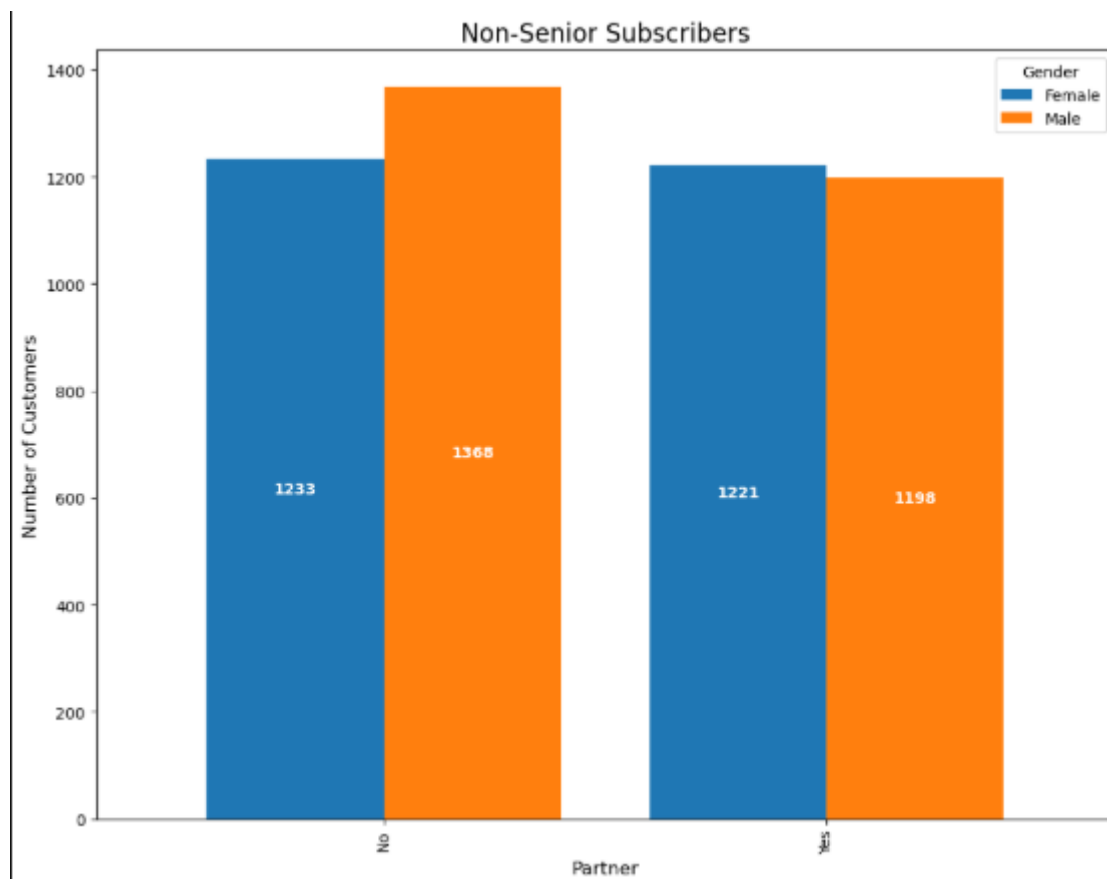


Figure 5: Non-senior subscribers with and without partners

Summaries

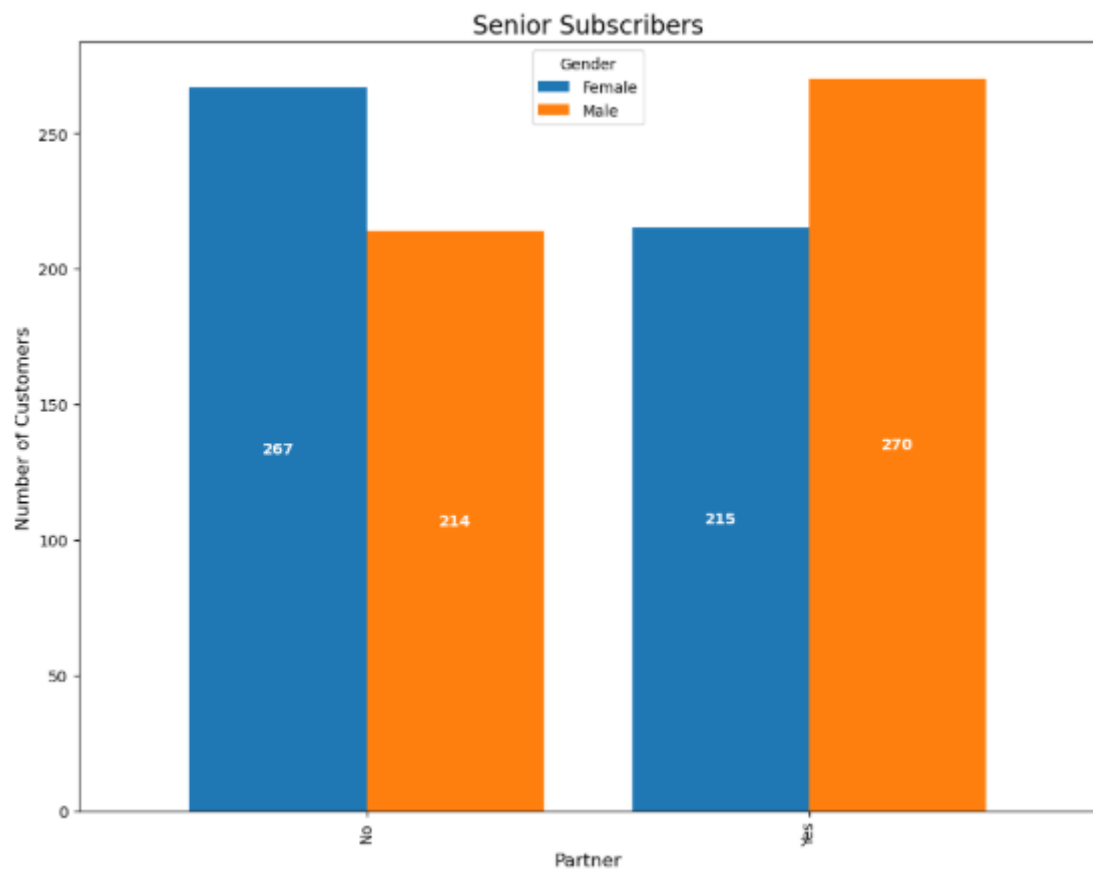


Figure 6: Senior subscribers with and without partners

The graphs above provides a further breakdown of the different subscriber demographic of the Telecom users.

We can see that roughly there is a similar ratio of those with partners and those without for both seniors and non-seniors regardless of gender.

Summaries

Additional Summaries

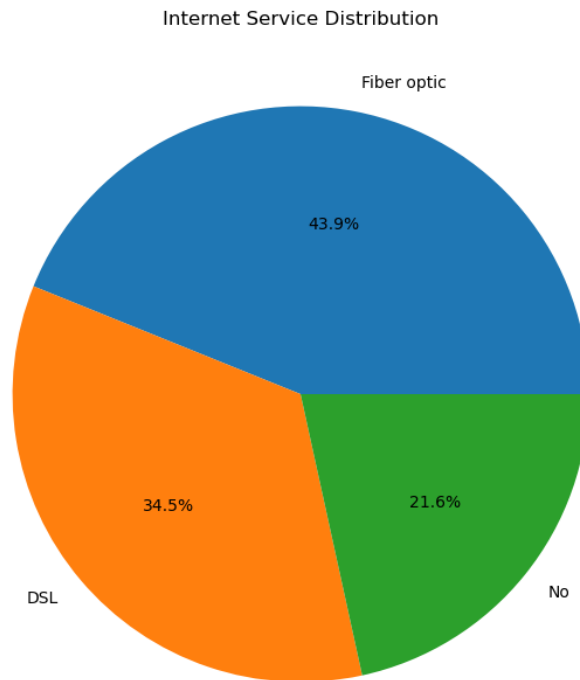


Figure 7: The pie chart above illustrates the internet distribution of those with Fibre optic, DSL or without.

We can see that a majority of telecom users use Fiber optic internet service

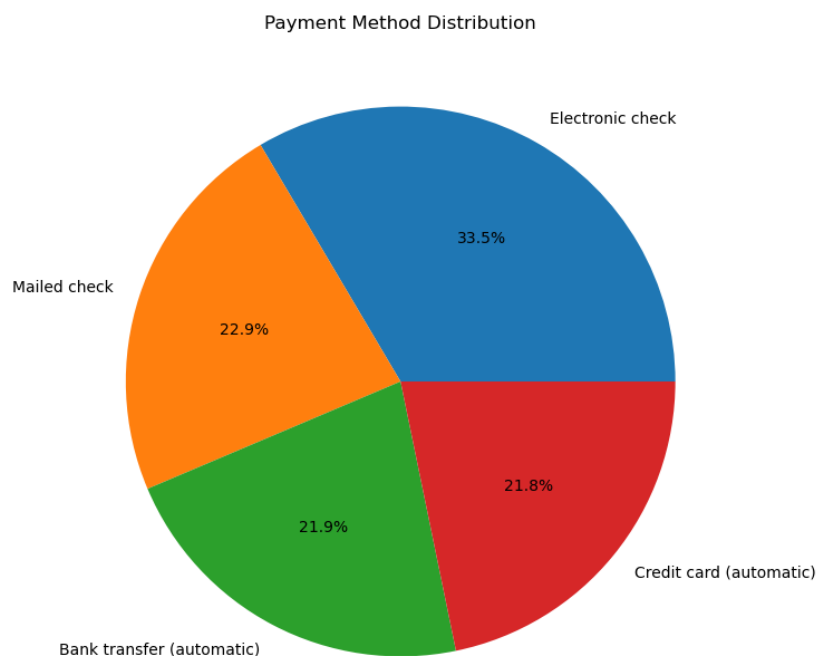


Figure 8: This pie chart provides the percentage of the varied payment methods of the telecom users within the dataset.

From what we can observe, the payment methods have similar distributions with electronic check being the most preferred.

Summaries

Additional Summaries: Churn Rate

Churn refers to the rate at which customers stop interacting with a company after a given period of time.

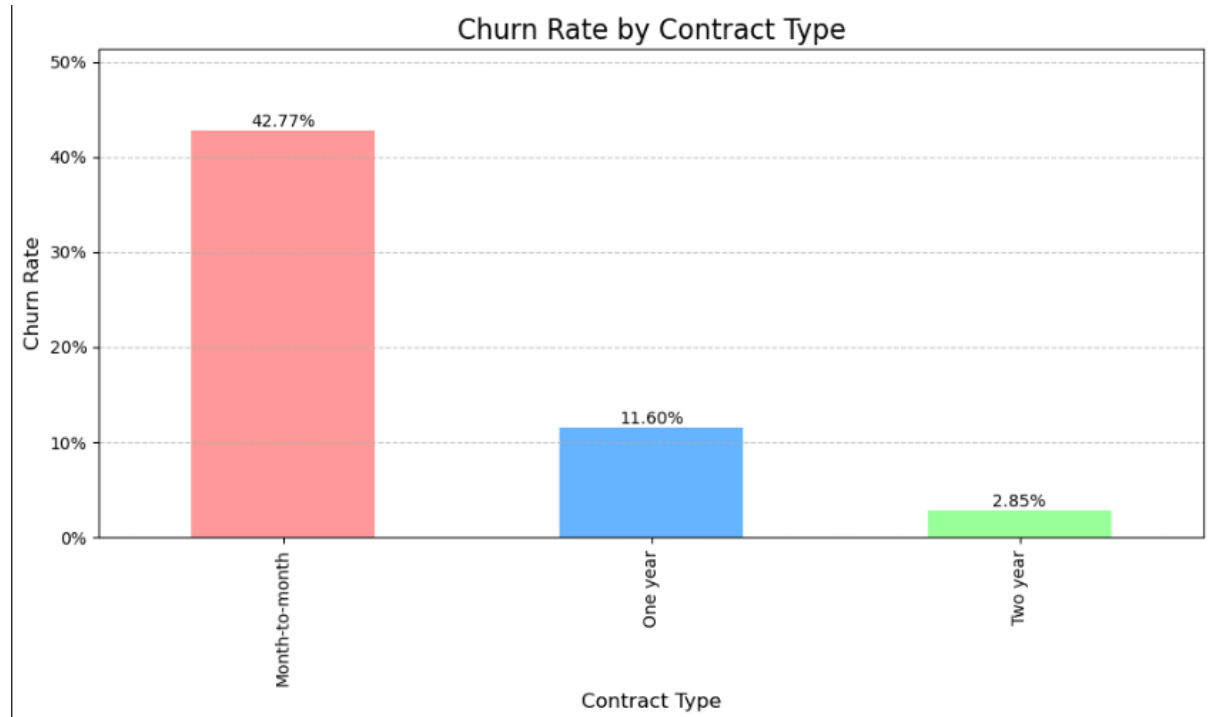


Figure 9: The graph indicates the churn rate based off the contract type.

Based off the table there is a trend where shorter contract led to more churn. This suggests that customers that have shorter contract times are more prone to switching and customers with longer contracts are more likely to be loyal customers and are unlikely to change to a different company.

Summaries

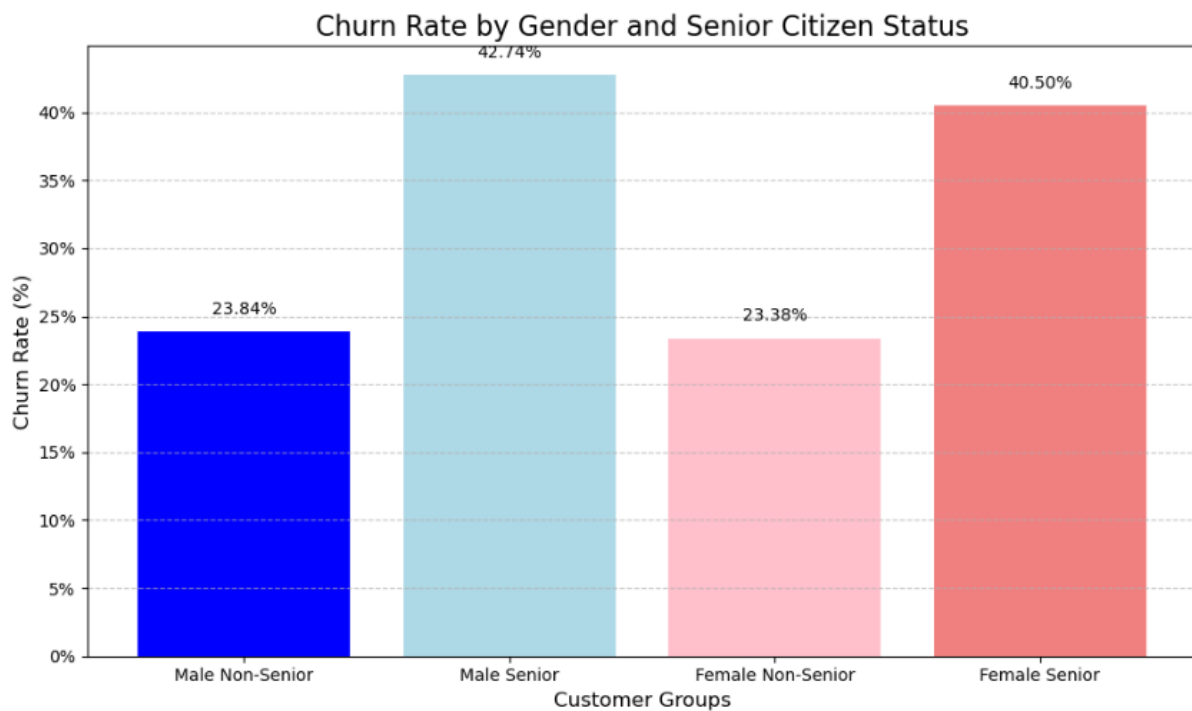


Figure 10: This Graph displays the churn rate based on factors of gender and senior citizenship

Looking at this graph there are two main things that are identifiable, the difference in churn rate comparing gender. Though minimal in differences, male subscribers have a slightly higher churn rate in comparison to female subscribers.

On the other hand, the other observable analysis is the difference based on senior status, where there is a considerably higher percentage churn rate found in those with senior status than those without. About 17-20 % difference.

To compare senior males, have churn rate of 42.74% whereas non-senior males have a churn rate of 23.84% this is a difference of 18.9% which is considerable.

Summaries

The two tables below display the distribution in percentages of both senior and non-senior subscribers based on their contract type. The values of the graphs are based on the ratio of senior and non-senior respectively.

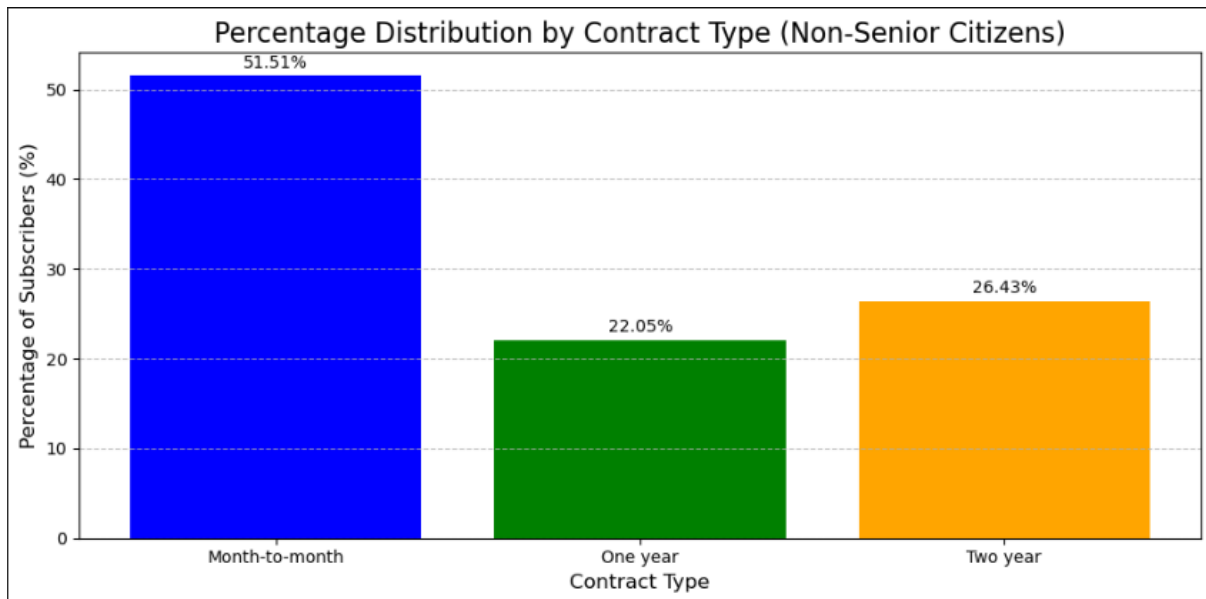


Figure 11:

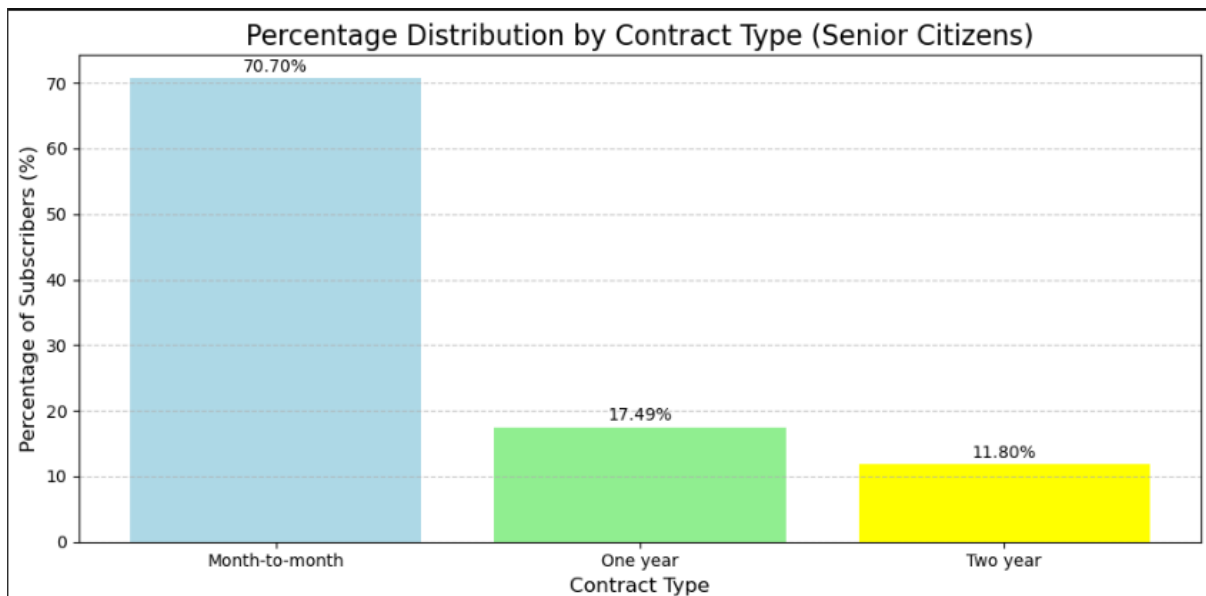


Figure 12:

Viewing the graphs, we can identify that both seniors and non-seniors prefer to contract month-to-month compared to other contract types. We can see that a much higher percentage of seniors prefer month-to-month contracts with a ratio nearly 20% comparatively to non-seniors.

In contrast there is a greater ratio of 'One year' and 'Two year' contracts for non-seniors.

Summaries

The table below displays the churn rate based off the contract type and senior status

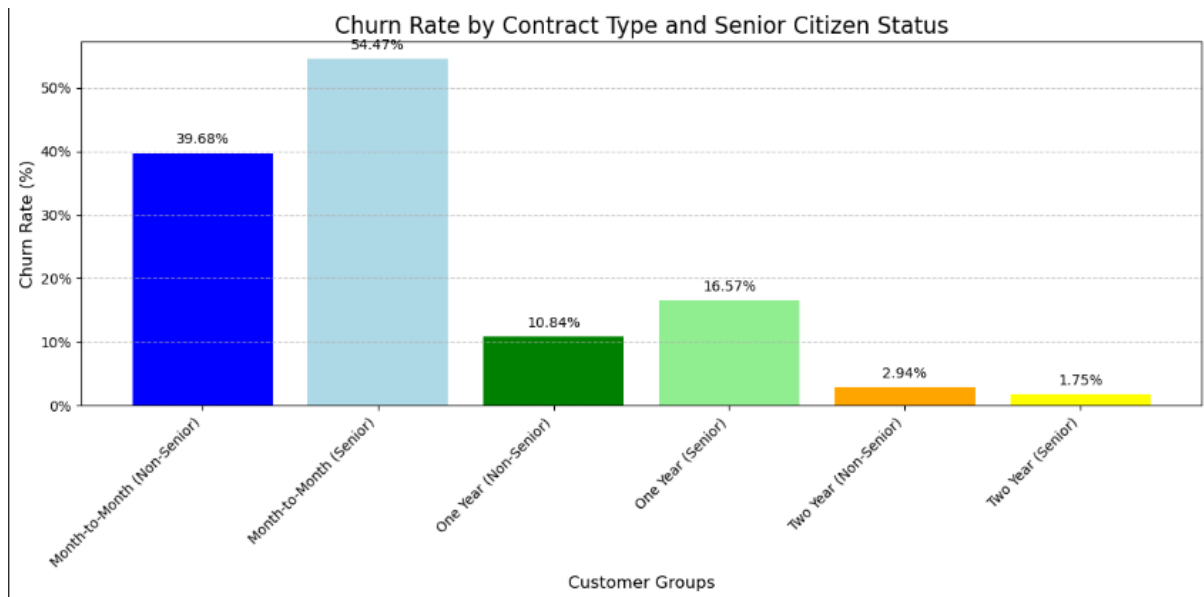


Figure 13:

From the combination of contract type and senior citizen status, we can break down from the previous graph of just contract and churn rate along with the other summaries for senior and non-senior status.

Just from this bar graph we can generally see that shorter contracts and senior subscribers tend to have a higher churn rate. Based on the previous summaries that support this, we can presume the various reason this might be the case.

Contract Type

Typically for shorter contracts they tend to be more flexible and temporary. This is because they are not locked into a long-term commitment, this could lead to more switching if there are better deal with other providers.

Whereas in longer contracts, customers may feel more invested in the service due to perceived cost savings or benefits leading to lower churn rates. They are likely to stick around more to get the full value of the contract

Senior Citizen Status

Some of the main reasons for higher churn rate for senior citizens might be because of changing circumstances and needs. As people age their needs and preferences may change, for example, during retirement many seniors are on fixed income, so they are more price-sensitive and tend to look for better deals elsewhere. On the other hand, some seniors may feel like their current telecom service does not meet their requirement any longer and might look elsewhere as a result.

Summaries

Additional Summaries: Monthly Charges based on Contract Type

The graph below is a box plot of the monthly charges based on the contract type

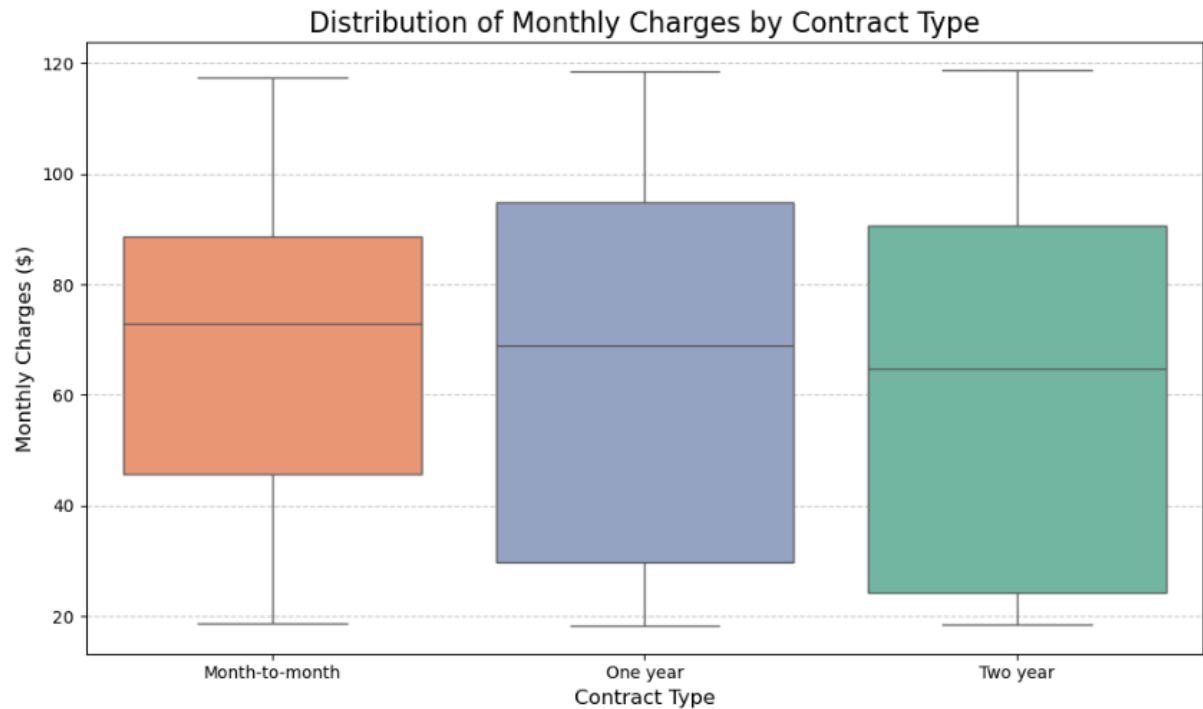


Figure 14:

The table below displays the exact values of the box plots above

Statistics of Monthly Charges by Contract Type:					
	Min	Q1	Median	Q3	Max
Contract					
Month-to-month	18.75	45.75	72.95	88.75	117.45
One year	18.25	29.71	69.00	94.80	118.60
Two year	18.40	24.15	64.80	90.65	118.75

Figure 15

Overall, the maximum and minimum monthly charges between the three different contract types are relatively similar.

From the month-to-month we can see that there is lower dispersion of per monthly charges within the interquartile range this means there is a lower range but does suggest the 25% in the third quartile cost between 72.95-88.75. Which could indicate higher costs on average.

On the other hand, with the One year and Two year contract the box plot has the interquartile range generally increase downwards where the lower quartile cost less monthly. This could indicate that overall, longer contract tend to cost less monthly.

Summaries

Additional Summaries:

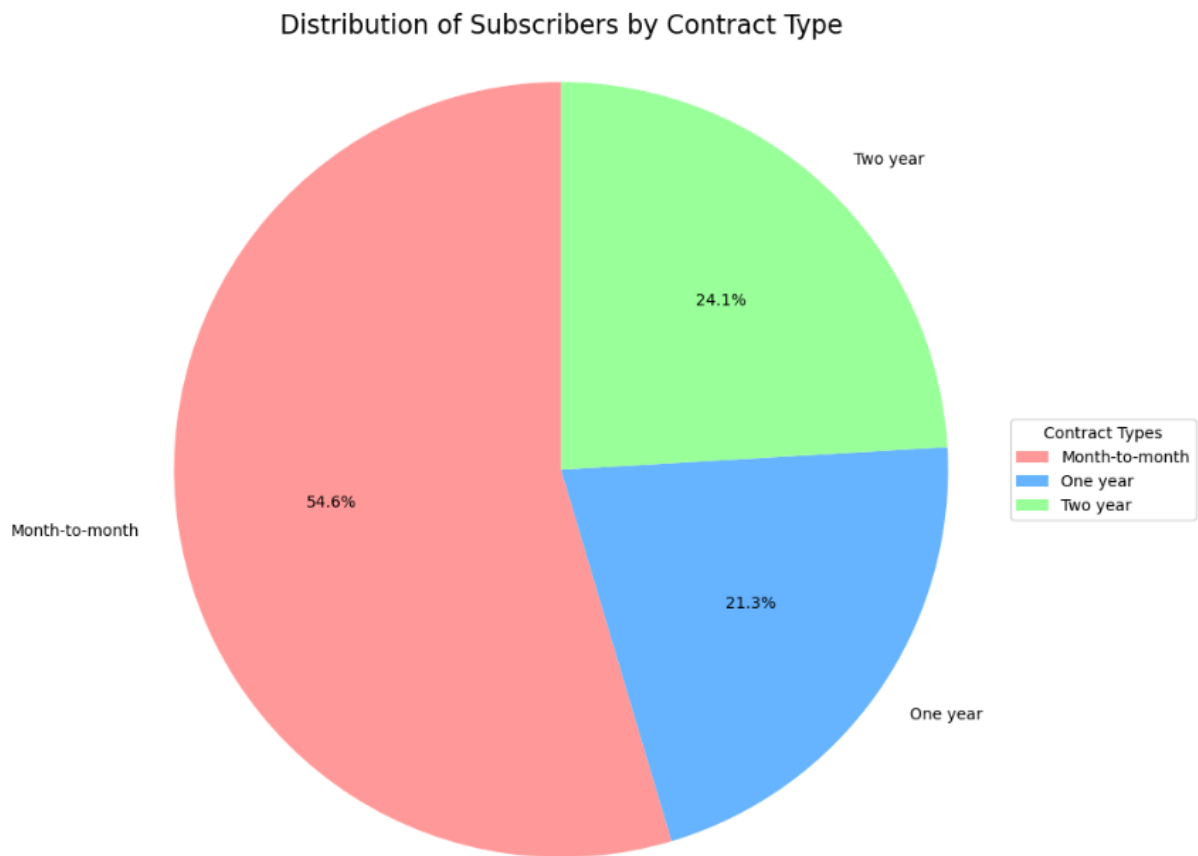


Figure 16: Distribution of total subscribers in each contract in percentages

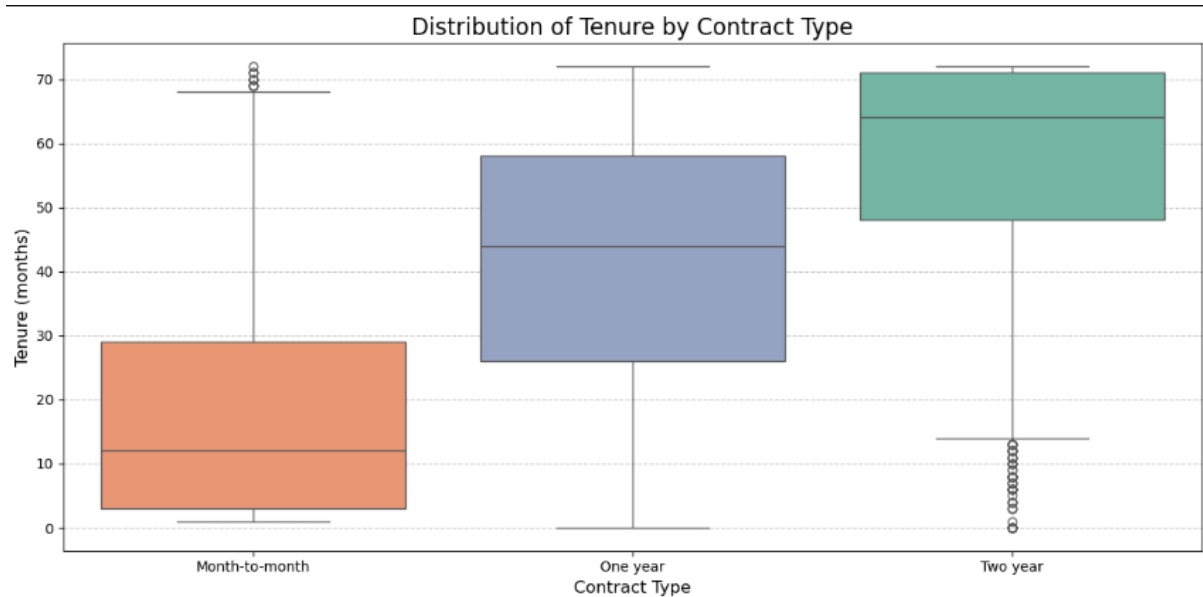


Figure 17

This box plot indicates that for the month-to-month contracts, typical tenure (in months) is more probable to be shorter where 75% of subscribers churn at just below 30 months with some outliers.

Summaries

On the other hand, the Two-year contract is more probable to have longer tenure (between 45-75 months) with outliers as well.

The One-year contract however, is in between with a more normal distribution compared to the other two plots which are very skewed.

Overall, this would be expected based on the earlier analysis of churn rate.

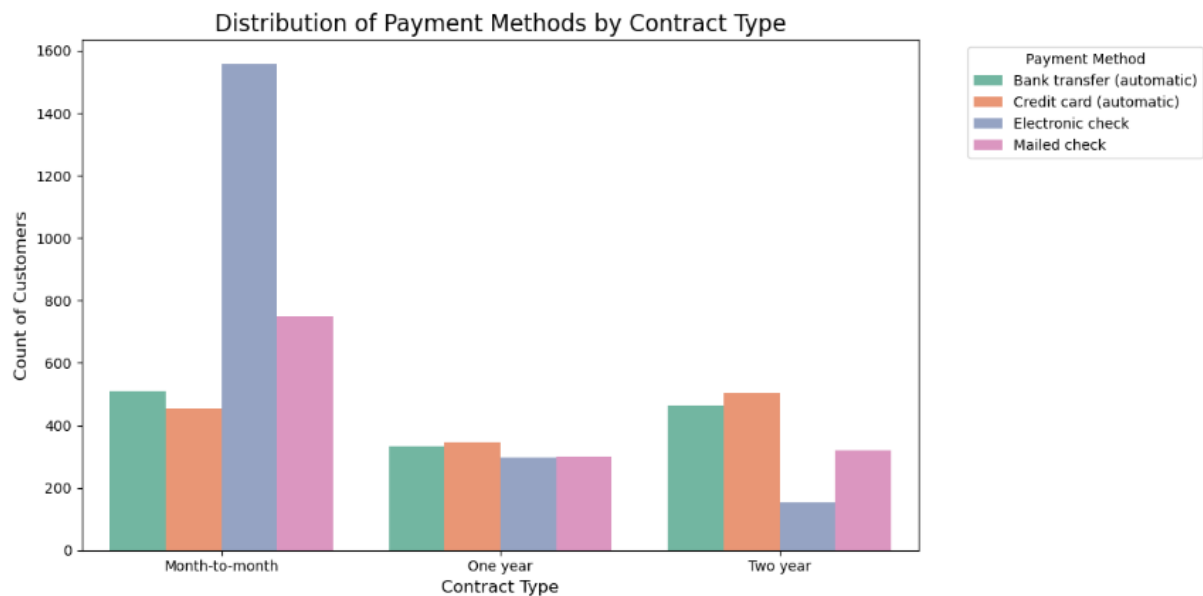


Figure 18: This table shows the distribution of payment methods based on contract type

This table indicates the subscriber's payment preference for each contract type.

For the month-to-month contract it shows that Electronic and Mailed checks are more popular, this is likely because it is easier to handle and control in the event that someone wants to change or move to a different provider.

Alternatively, for One-year and Two-year contracts Bank transfer and Credit card payments are slightly more popular. This is indicative to it being long term and people might prefer to do it automatically as it becomes more of a hassle to manually pay monthly for the duration.

Summaries

The two graphs below specify the average charges based on the type of internet and phone services

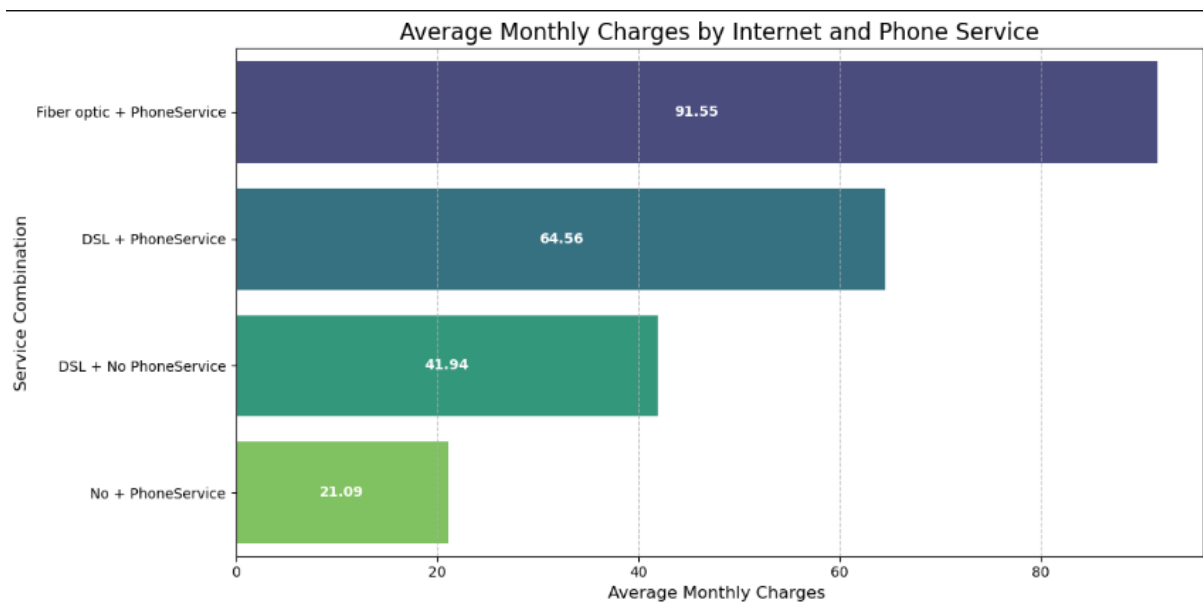


Figure 19: This graph shows the monthly charges based on the combination of different internets and whether subscribers have phone service

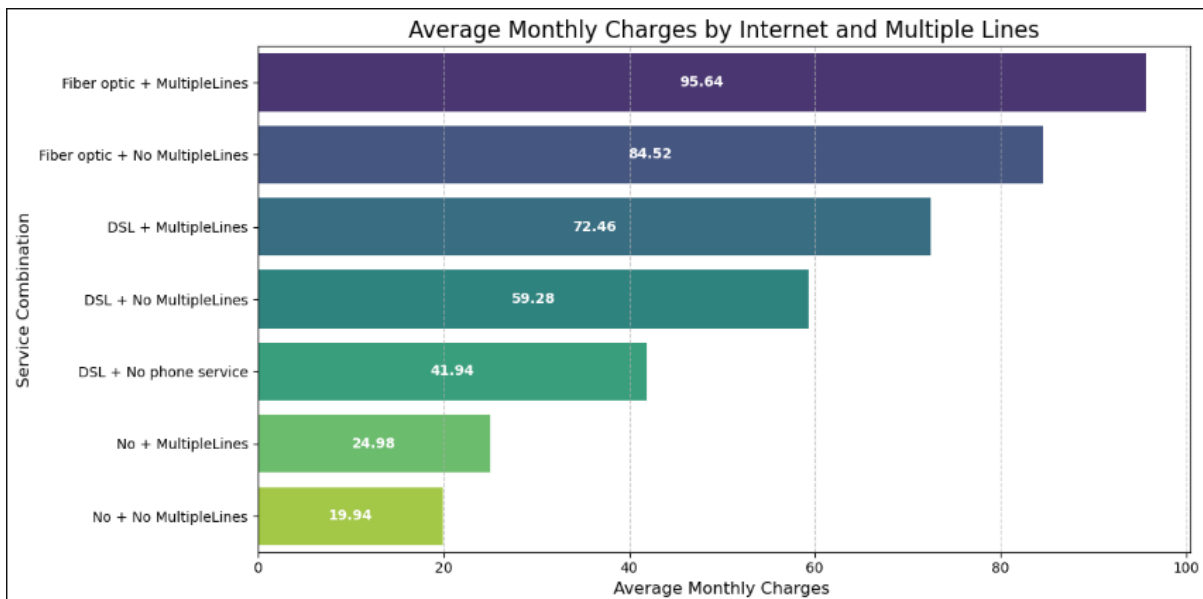


Figure 20: This graph similar to the previous figure shows different groupings of internet service and whether subscribers have multiple lines.

From the tables above we can see that different internet and phone services does affect the monthly charge for the telecom user.

We can identify that for internet the more expensive type is Fiber optic followed by DSL. We can make that comparison based on the first graph where both have phone service.

We can also observe that those with Fiber optic will have a phone service.

We can conclude that the inclusion of different internet types and phone service will significantly affect the monthly charges overall.